

Beyond Epsilon: A Principled QIF Framework for Local Differential Privacy

Ramon G. Gonze

UFMG / Institut Polytechnique de Paris

Natasha Fernandes

Mcquarie University

Héber H. Arcolezi

ETS Montréal

Catuscia Palamidessi

Institut Polytechnique de Paris / Inria

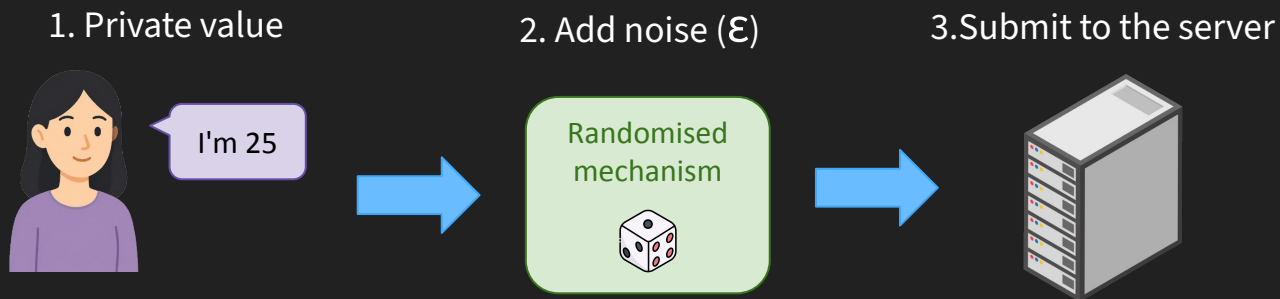
Nataliia Bielova

Inria Centre at University Côte d'Azur

39th IEEE Computer Security Foundations Symposium | July 26, 2026

Motivation

- Local Differential Privacy (LDP)



- Widely used for frequency estimation in large datasets
- How to compare LDP mechanisms?

Motivation

ϵ

*Is it enough to
compare mechanisms?*



Bounds worst-case distinguishability

Ranking by utility for a given budget ϵ

Fail to account: Strength of protocols
against diverse attacker models.



Data
reconstruction

Re-identification



*What does it mean for one LDP protocol to
be **strictly more private** than another?*

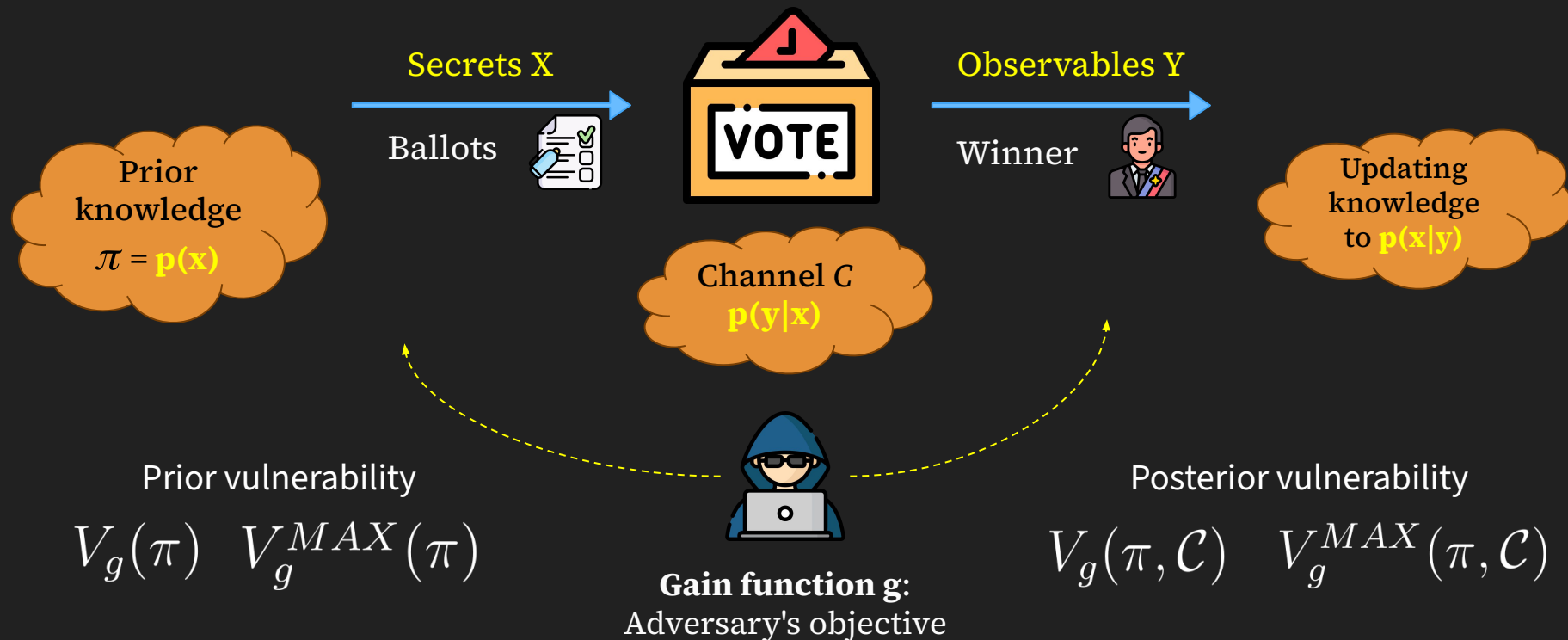


Contributions

1. Unified QIF-based formal model for comparing 7 LDP mechanisms.
 - Metric-independent
 - Refinement relations
2. Refinement: Is A strictly more private than B ?
3. Bridge LDP and QIF literatures.
4. Correct an error in the analysis of Local Hashing (LH) vulnerability.

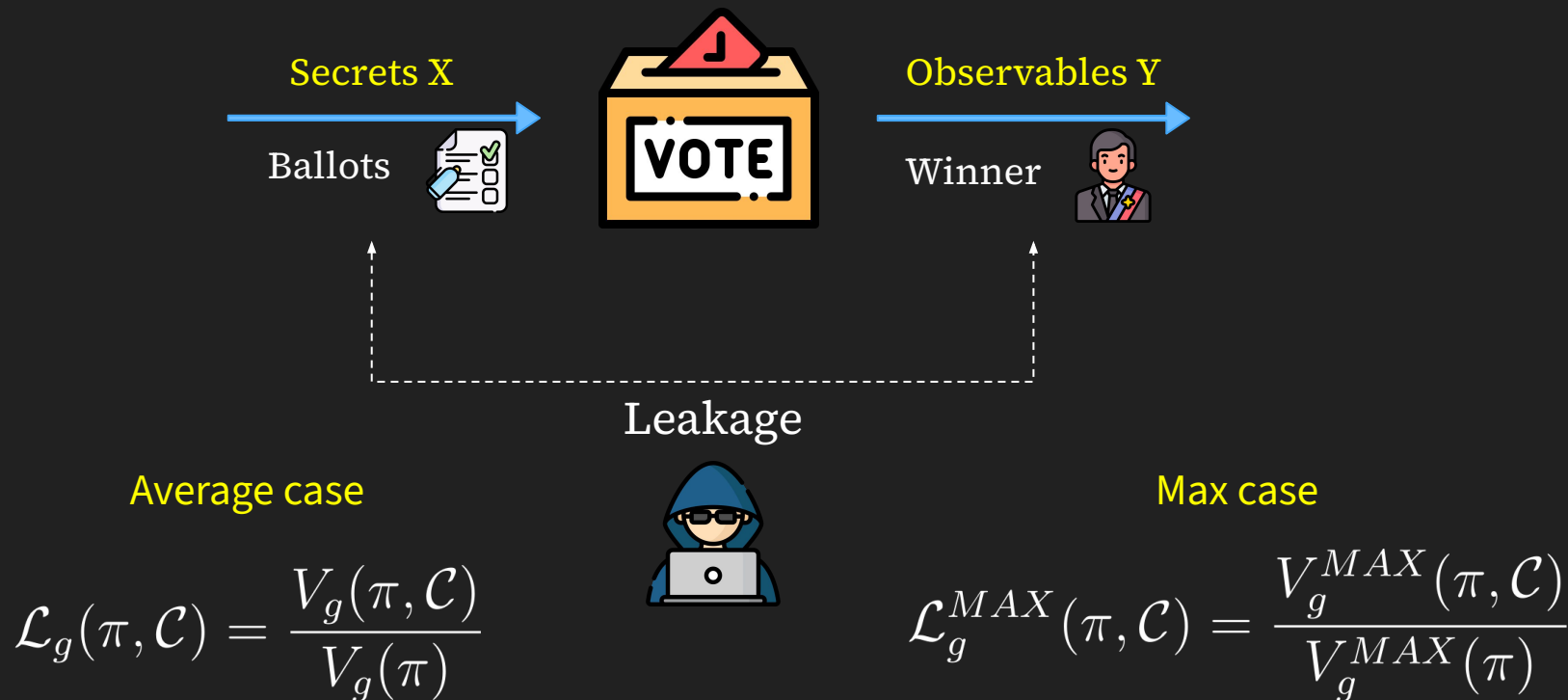
Quantitative Information Flow (QIF)

An election with candidates and voters.



Quantitative Information Flow (QIF)

An election with candidates and voters.



Capacity

Average case capacity [1]

$$\begin{aligned}\mathcal{ML}(\mathcal{C}) &= \max_{\pi, g} \mathcal{L}_g(\pi, \mathcal{C}) \\ &= \sum_y \max_x \mathcal{C}_{x,y}\end{aligned}$$

$$\text{Bayes gain} = \begin{cases} 1 & , \text{ if guess} = \text{secret} \\ 0 & , \text{ otherwise} \end{cases}$$

Max-case capacity [2]

$$\begin{aligned}\mathcal{ML}^{MAX}(\mathcal{C}) &= \max_{\pi, g} \mathcal{L}_g^{MAX}(\pi, \mathcal{C}) \\ &= e^\epsilon\end{aligned}$$

[1] M. S. Alvim, K. Chatzikokolakis, C. Palamidessi, and G. Smith, “Measuring information leakage using generalized gain functions,” in 2012 IEEE 25th Computer Security Foundations Symposium, pp. 265–279, IEEE, 2012.

[2] N. Fernandes, A. McIver, and P. Sadeghi, “Explaining ϵ in local differential privacy through the lens of quantitative information flow,” in 2024 IEEE 37th Computer Security Foundations Symposium (CSF), pp. 419–432, 2024.

Refinement

Does **A** leak more information than **B**?

- Regardless the prior knowledge
- Regardless the objective (gain function)

$$A \sqsubseteq B$$

B refines **A**  **A** *always leaks* more than **B**

f -differential privacy

- Characterizes privacy guarantees in terms of hypothesis testing.
- Supports comparison of mechanisms



H_0 : target **not in**
the data



H_1 : target
in the data

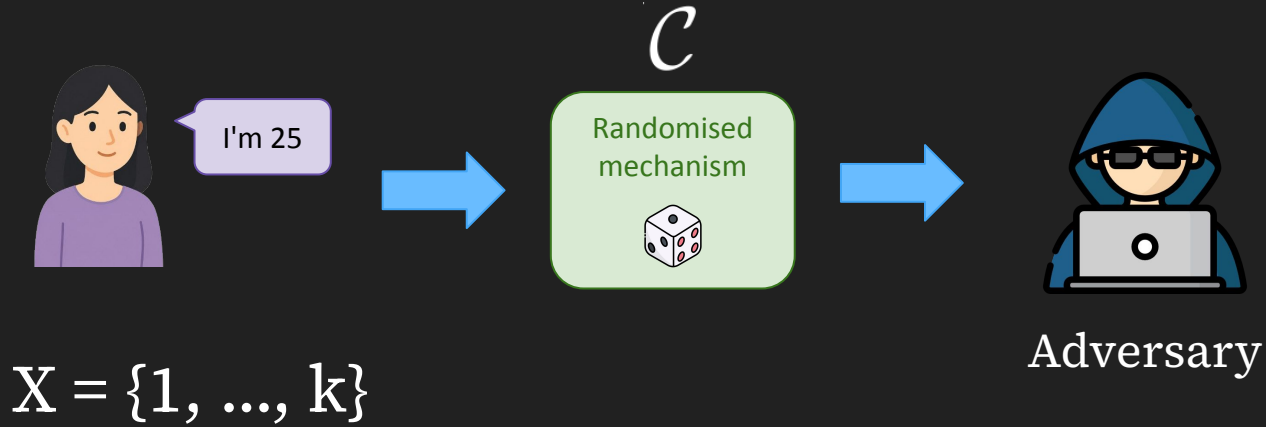


Type-I (α) error

Type-II (β) error

Trade-off function $T_M(\alpha) = \beta$

Formal model



$$\mathcal{ML}(\mathcal{C})$$

Average-case capacity

Leakage measures

Adversarial Success Rate (ASR) :

Ratio of clients whose true value is correctly predicted.

$$\text{ASR}(\mathcal{C}) = V_1(\pi, \mathcal{C}) \quad \longrightarrow \quad \text{ASR}(\mathcal{C}) = \mathcal{ML}(\mathcal{C})/k$$

Min-entropy leakage:

X = R.V. Set of secrets

Y = R.V. Channel's set of outputs

[1]

$$\mathcal{L}_\infty(X | Y) = H_\infty(X) - H_\infty(X | Y) \quad \longrightarrow \quad \mathcal{L}_\infty(X | Y) = \log(\mathcal{ML}(\mathcal{C}))$$

[1] M. S. Alvim, K. Chatzikokolakis, C. Palamidessi, and G. Smith, "Measuring information leakage using generalized gain functions," in 2012 IEEE 25th Computer Security Foundations Symposium, pp. 265– 279, IEEE, 2012.

LDP Protocols and their leakages

How many times did you say "I'll do it tomorrow" this week?

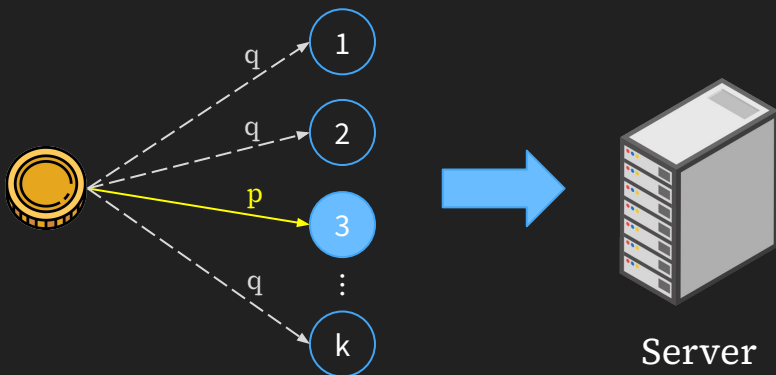
$$Y = \{1, \dots, k\}$$



Ana

3 times

GRR: Generalized Randomised Response



$$\mathcal{ML}(\mathcal{R}^\epsilon) = pk = \frac{ke^\epsilon}{e^\epsilon + k - 1}$$

LDP Protocols and their leakages

How many times did you say "I'll do it tomorrow" this week?

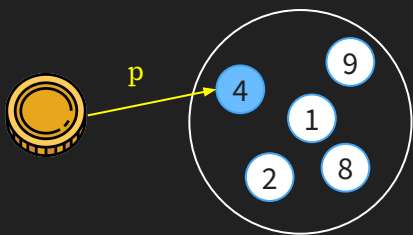
$$\mathcal{Y} = \{1, \dots, k\}$$



Ana

3 times

SS: Subset Selection



Set size $\omega = 5$



Server

$$\mathcal{ML}(\mathcal{SS}^\varepsilon) = \begin{cases} \frac{kp}{\binom{k-1}{\omega-1}} & , \text{ if } \frac{p}{\binom{k-1}{w-1}} > \frac{1-p}{\binom{k-1}{w}} \\ \frac{k(1-p)}{\binom{k-1}{\omega}} & , \text{ otherwise} \end{cases}$$

LDP Protocols and their leakages

How many times did you say "I'll do it tomorrow" this week?

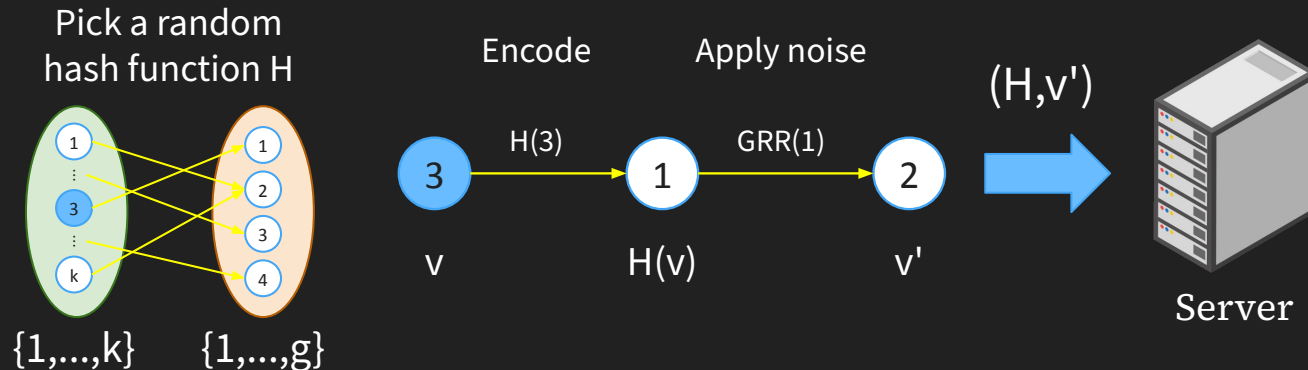
$$Y = \{1, \dots, k\}$$



Ana



Local Hashing protocols (**BLH** and **OLH**)



LDP protocols and their leakages

Local Hashing Protocols (**BLH** and **OLH**)

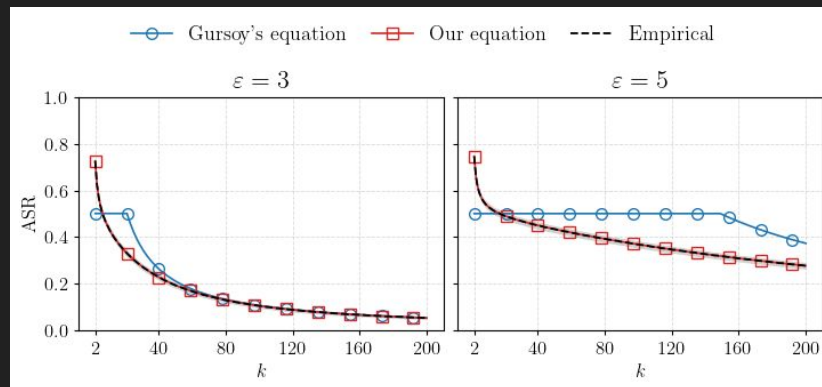
ASR presented in [3]:

$$\frac{e^\varepsilon}{(e^\varepsilon + g - 1) \cdot \max\left\{\frac{k}{g}, 1\right\}}$$



Our correction:

$$\text{ASR}(\mathcal{LH}^\varepsilon) = \frac{e^\varepsilon g^k + (g - 1)^k (1 - e^\varepsilon)}{(e^\varepsilon + g - 1)(kg^{k-1})}$$



[3] M. Emre Gursoy, L. Liu, K.-H. Chow, S. Truex, and W. Wei, "An adversarial approach to protocol analysis and selection in local differential privacy," IEEE Transactions on Information Forensics and Security, vol. 17, pp. 1785–1799, 2022.

LDP Protocols and their leakages

How many times did you say "I'll do it tomorrow" this week?

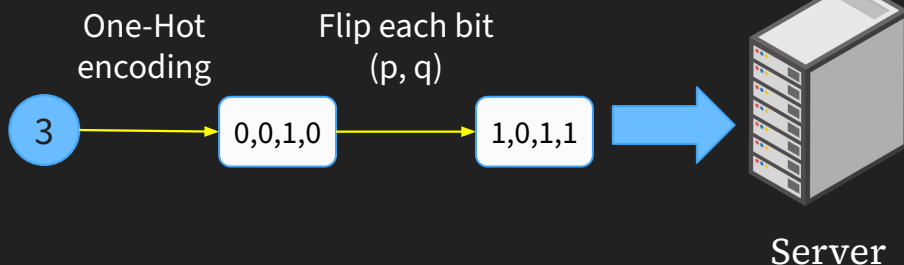


Ana

3 times

$$\mathcal{Y} = \{1, \dots, k\}$$

Unary Encoding protocols (**SUE** and **OUE**)



$$\mathcal{ML}(\text{SUE}^\varepsilon) = 1 + p^{k-1}(1 - 2p)/(1 - p)$$

$$\mathcal{ML}(\text{OUE}^\varepsilon) = \frac{1}{2q} \left((1 - q)^{k-1}(2q - 1) + 1 \right)$$

LDP Protocols and their leakages

How many times did you say "I'll do it tomorrow" this week?

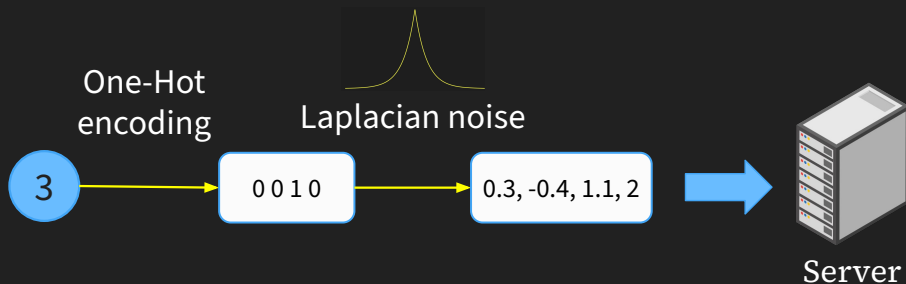


Ana

3 times

$$\mathcal{Y} = \{1, \dots, k\}$$

THE: Threshold with Histogram Encoding



$$\mathcal{ML}(\mathcal{T}^\varepsilon) = (1 - q)^{k-1} \left(1 - \frac{p}{q} \right) + \frac{p}{q}$$

2x2 channels

Define a 2x2 channel as

$$C = \begin{bmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{bmatrix}$$

→ Type-I error (pointing to α)

→ Type-II error (pointing to β)

SUE, OUE and THE



0 0 0 0 0 0 0 0 0 0 0 1 0

Lemma 1: Let **A** and **B** be 2x2 channels.

$$A \sqsubseteq B \implies A_{\text{hot}}^{\otimes k} \sqsubseteq B_{\text{hot}}^{\otimes k}$$

Bitwise refinement **implies** full-vector refinement.

Relation QIF and f -differential privacy

Lemma 2: Let $\mathbf{C}, \mathbf{D}$ be 2×2 channels with trade-off points (α_C, β_C) and (α_D, β_D) .

The following statements are equivalent:

1. $C \subseteq D$

2. $\frac{\beta_C}{1 - \alpha_C} \leq \frac{\beta_D}{1 - \alpha_D}$ and $\frac{1 - \beta_C}{\alpha_C} \geq \frac{1 - \beta_D}{\alpha_D}$

Refinement relations

1. $\text{SUE} \not\sqsubseteq \text{OUE}$ and $\text{OUE} \not\sqsubseteq \text{SUE}$

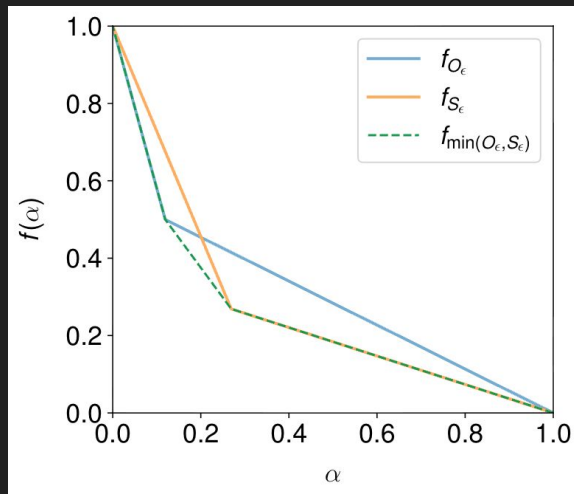
2. $\text{SUE} \sqsubseteq \text{THE}$

3. $\text{OUE} \sqsubseteq \text{THE}$ exactly when

$$\theta \geq \frac{2 \ln(e^\varepsilon + 1 + (e^{\varepsilon/2} - 1)(\sqrt{e^\varepsilon + 1})) - \varepsilon - 2 \ln 2}{\varepsilon}$$

A refinement for OUE and SUE

By the results in [1], there is a mechanism $M_\epsilon = S_\epsilon \min O_\epsilon$ which is the **anti-refinement** of OUE and SUE protocols, i.e., always leaks more information.



[1] N. Fernandes, A. McIver, and P. Sadeghi, “Composition theorems for f -differential privacy,” arXiv preprint arXiv:2512.21358, 2025.

Max-case refinement

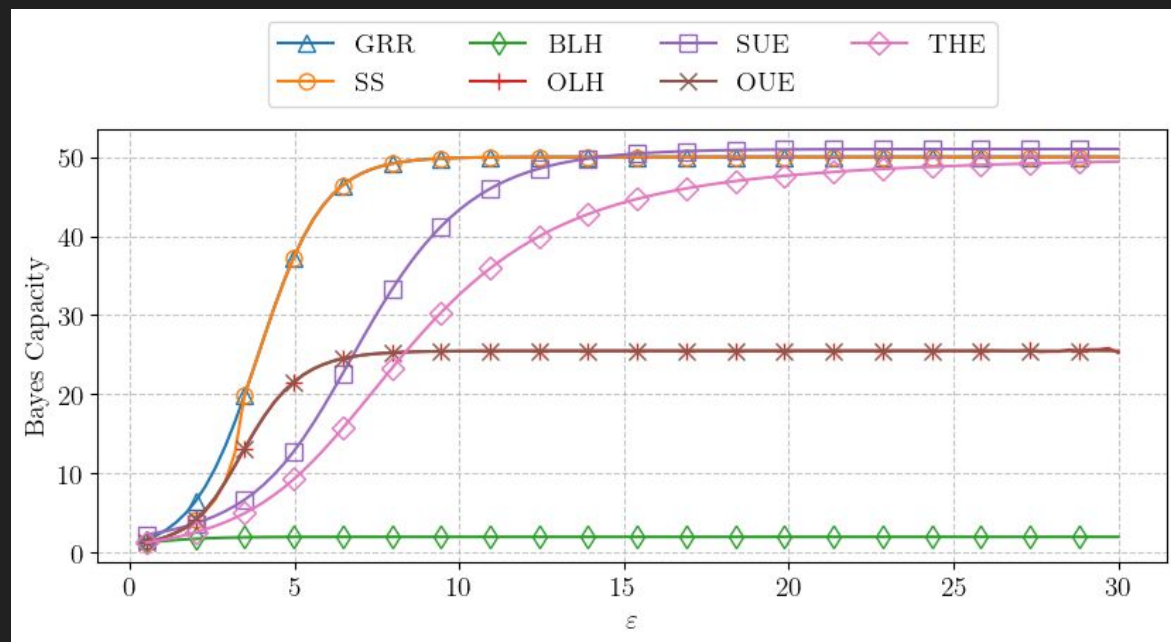
We show that average-case refinement coincides with max-case refinement.

$$A \sqsubseteq B \implies A \sqsubseteq^{MAX} B$$

Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

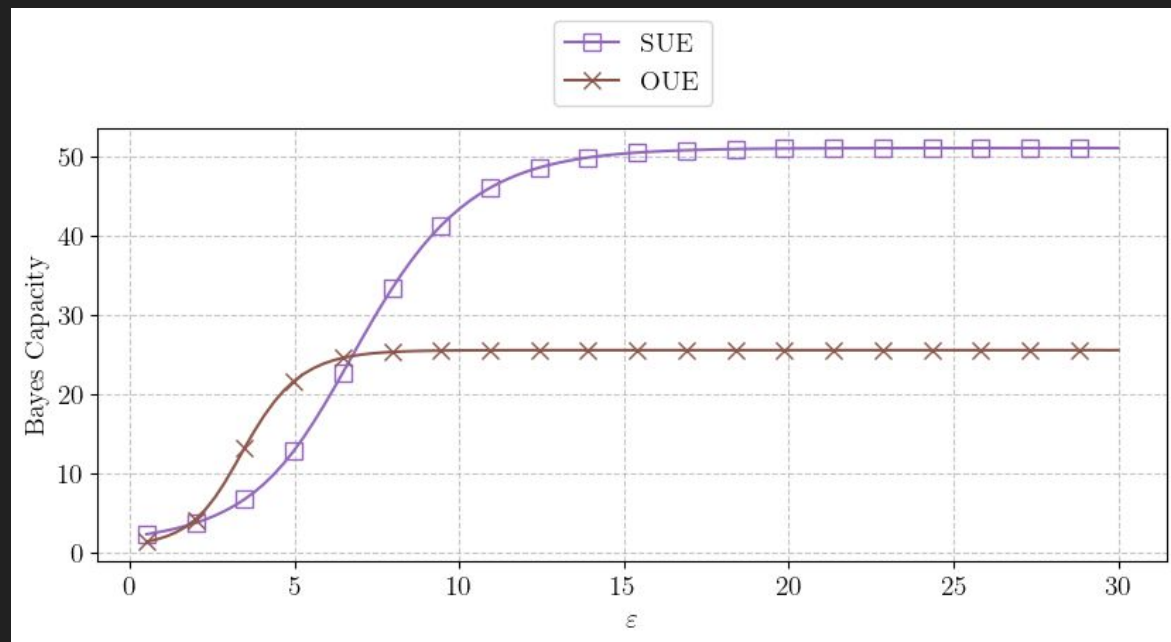
For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.



Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.

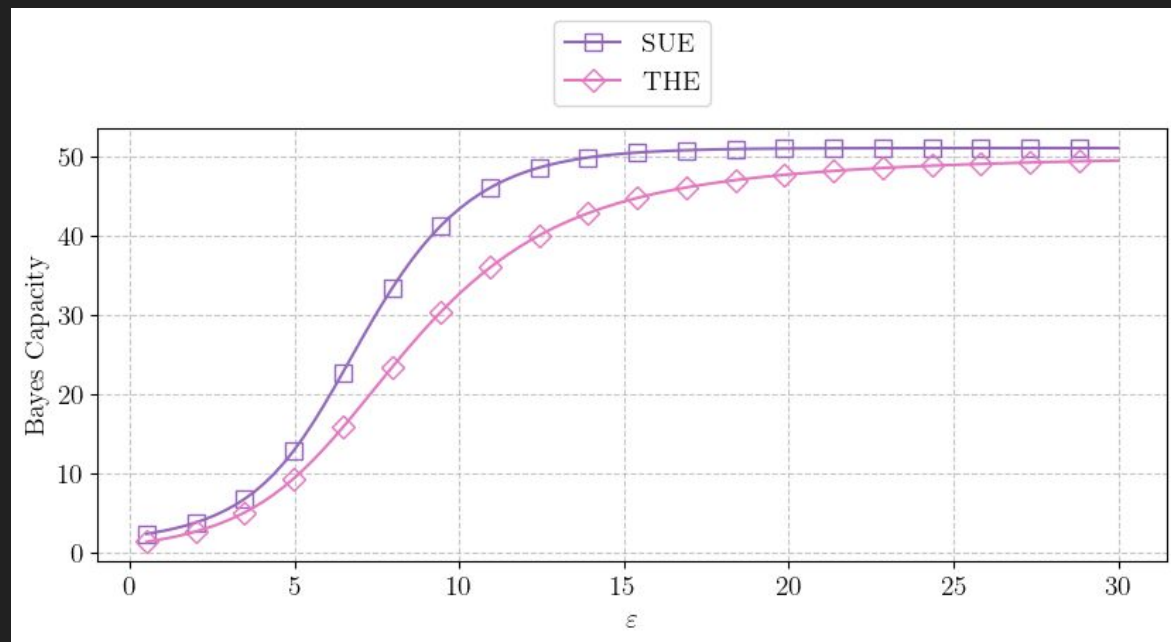


$SUE \not\subseteq OUE$ and
 $OUE \not\subseteq SUE$

Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.

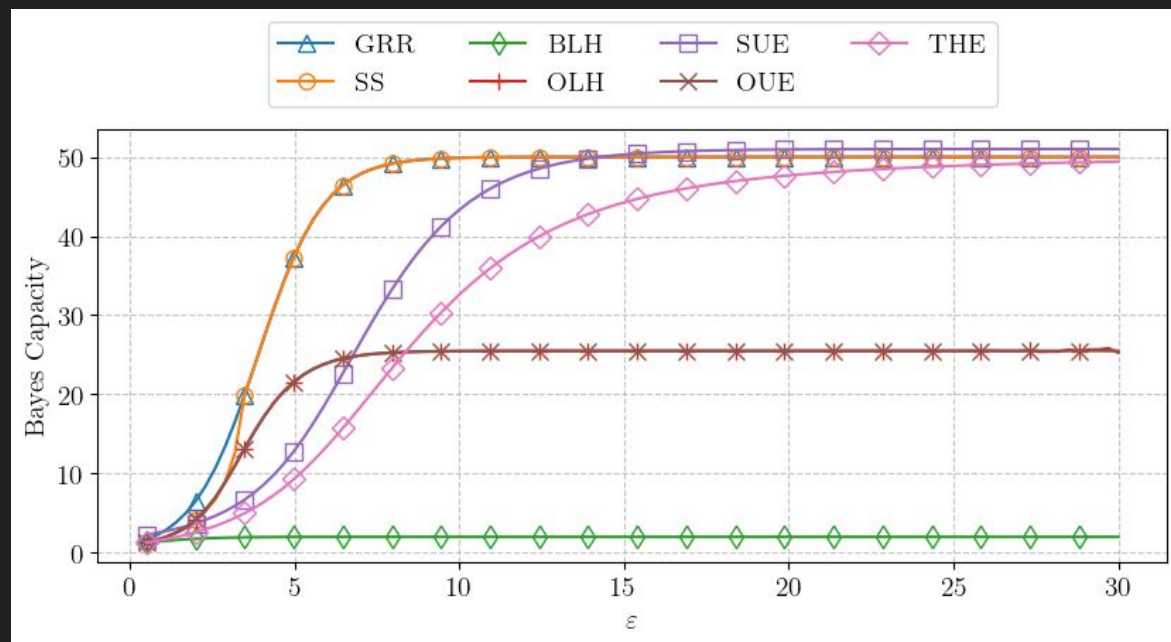


$$\text{SUE} \subseteq \text{THE}$$

Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

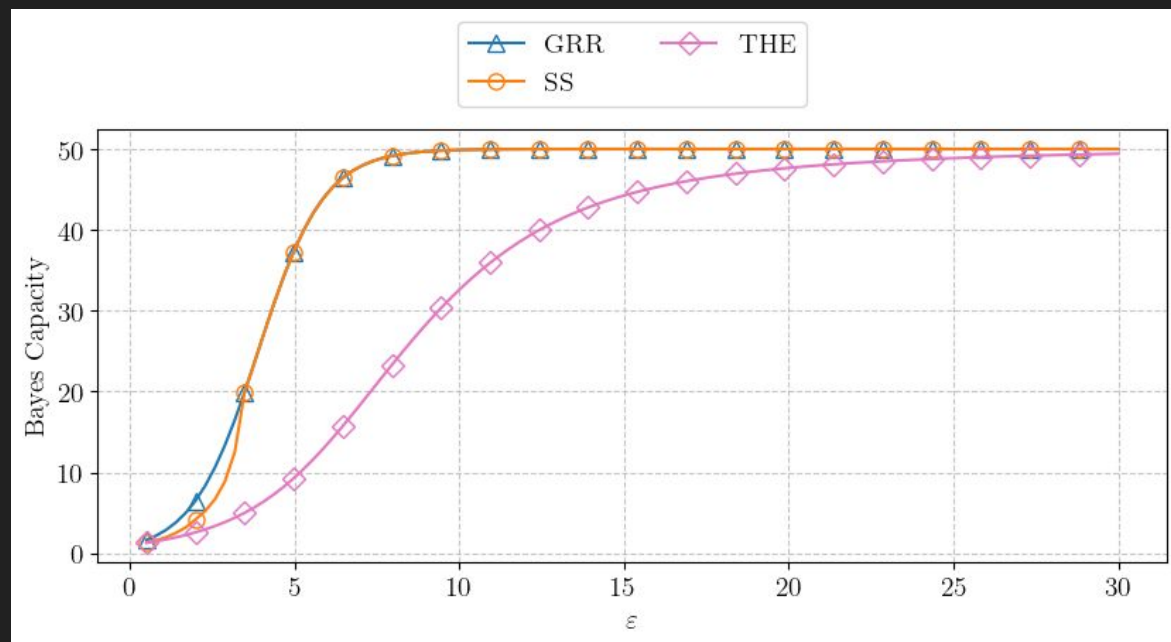
For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.



Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.

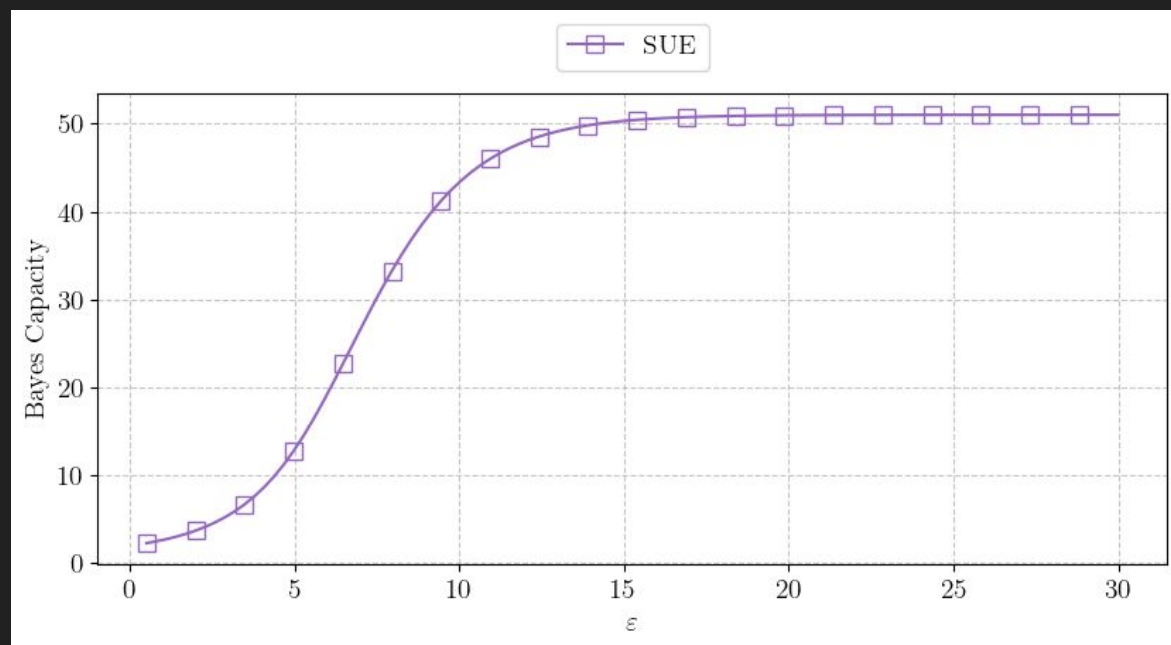


Converges to k

Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.

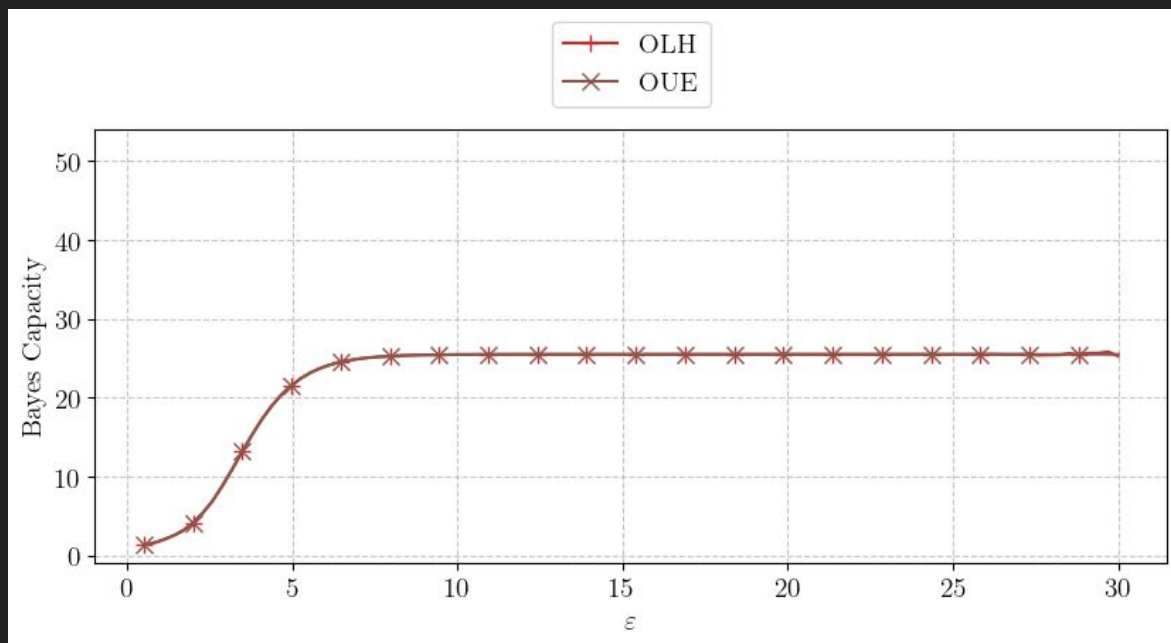


Converges to a value slightly higher than k

Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.

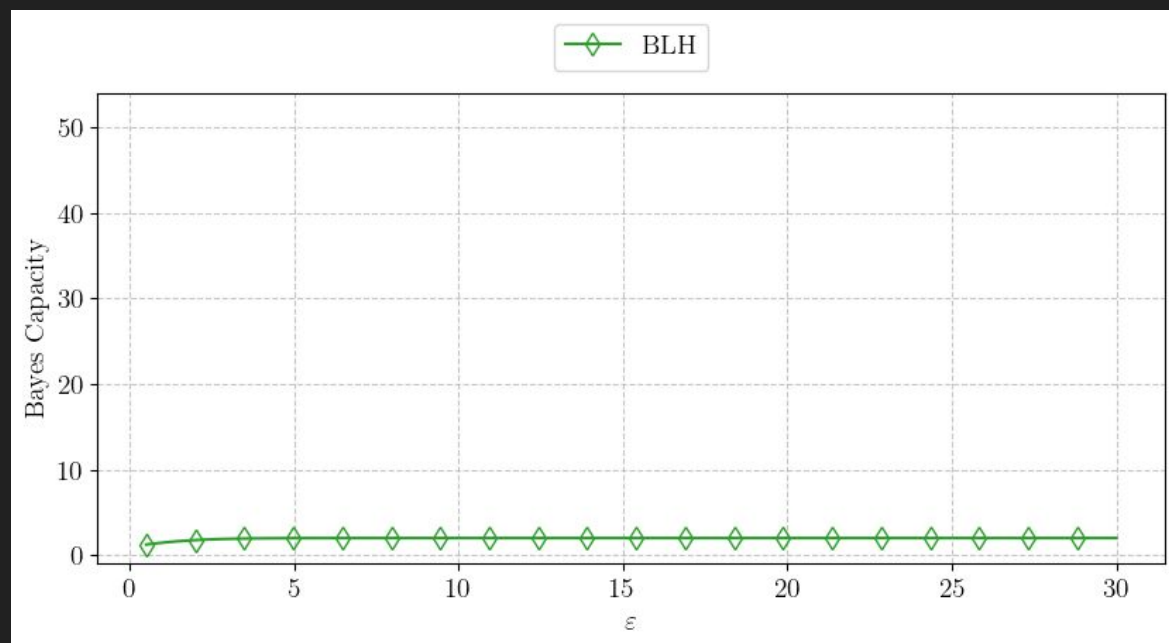


Converges to $k/2$

Experiments

Comparison of Bayes Capacity of all protocols, for $k=50$.

For THE, $\theta=3/4$, and all other protocol-specific parameters are optimal.



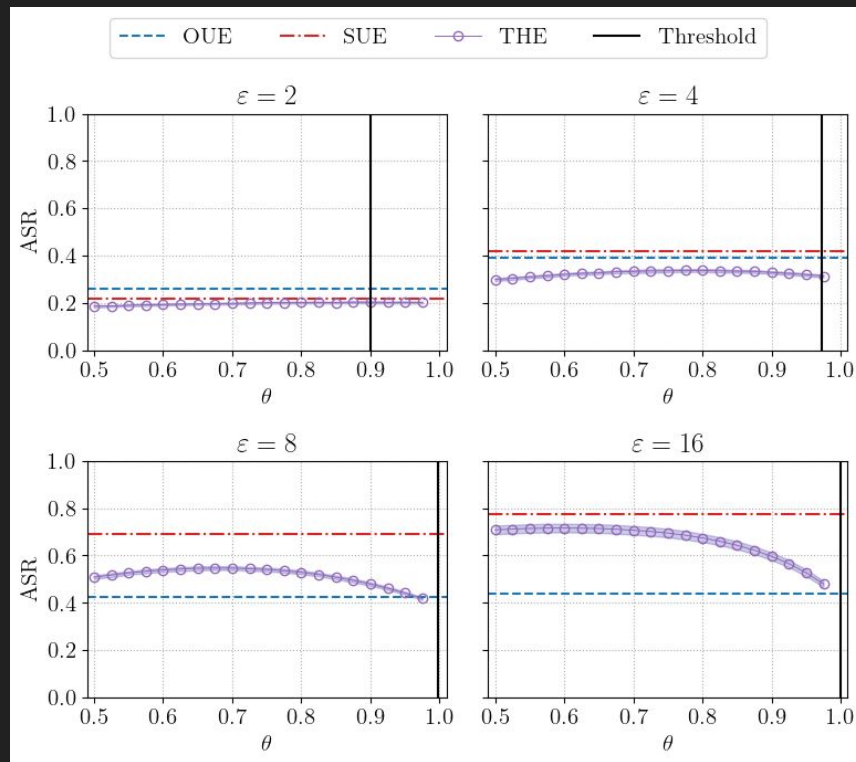
Converges to 2

Experiments

Kosarak dataset: Sequences of actions (clicks) performed by users while browsing a website, from a Hungarian news portal.

- 41,270 distinct actions
- 8kk actions performed by 990k users.

Comparison of ASR
Sample of 10,000 actions,
top 10 most frequent actions.

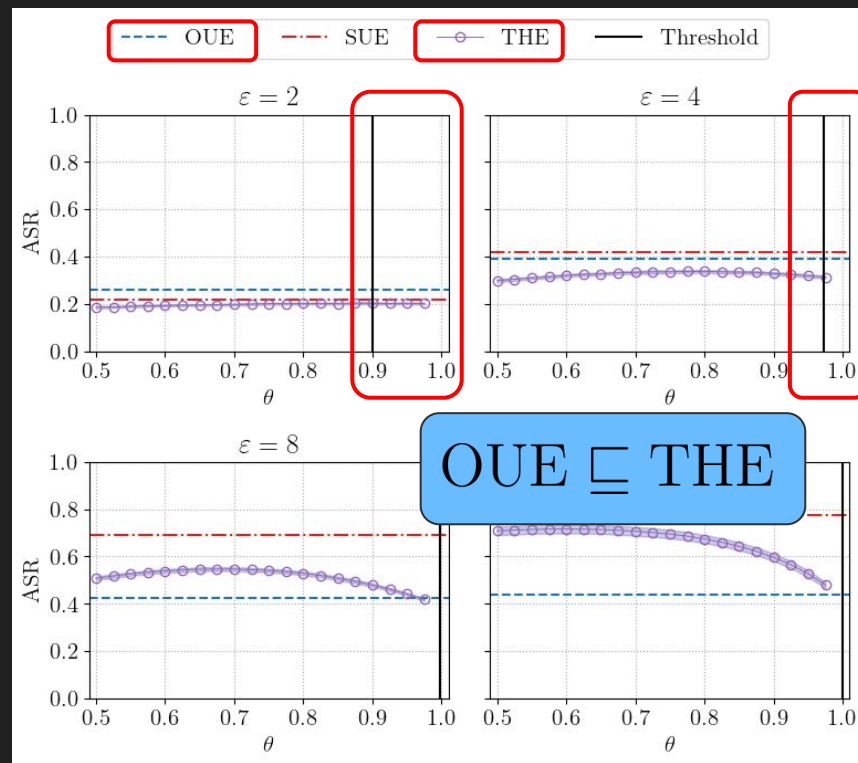


Experiments

Kosarak dataset: Sequences of actions (clicks) performed by users while browsing a website, from a Hungarian news portal.

- 41,270 distinct actions
- 8kk actions performed by 990k users.

Comparison of ASR
Sample of 10,000 actions,
top 10 most frequent actions.

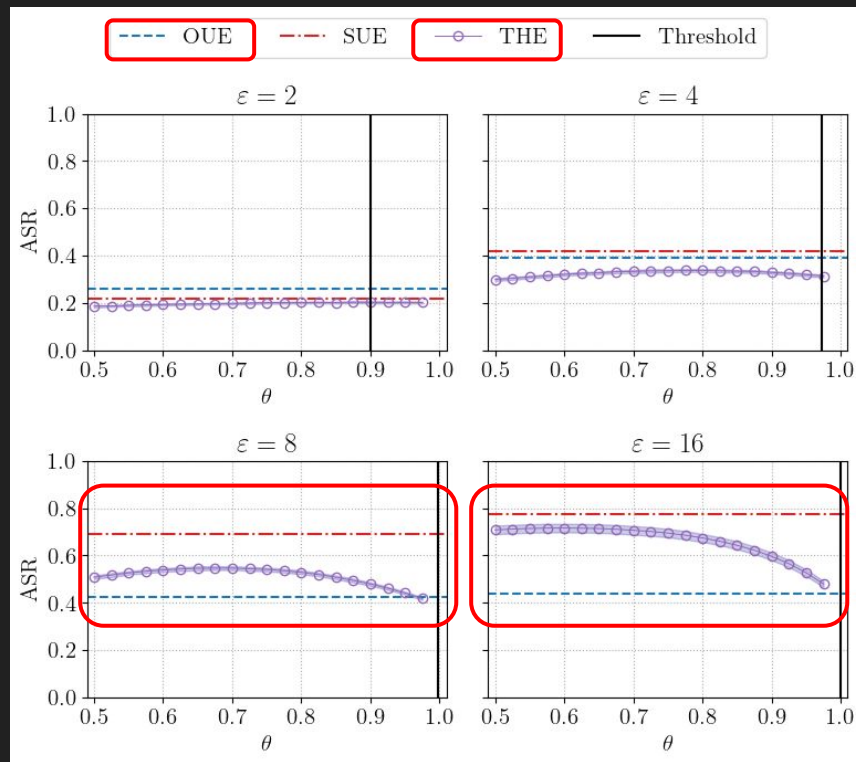


Experiments

Kosarak dataset: Sequences of actions (clicks) performed by users while browsing a website, from a Hungarian news portal.

- 41,270 distinct actions
- 8kk actions performed by 990k users.

Comparison of ASR
Sample of 10,000 actions,
top 10 most frequent actions.

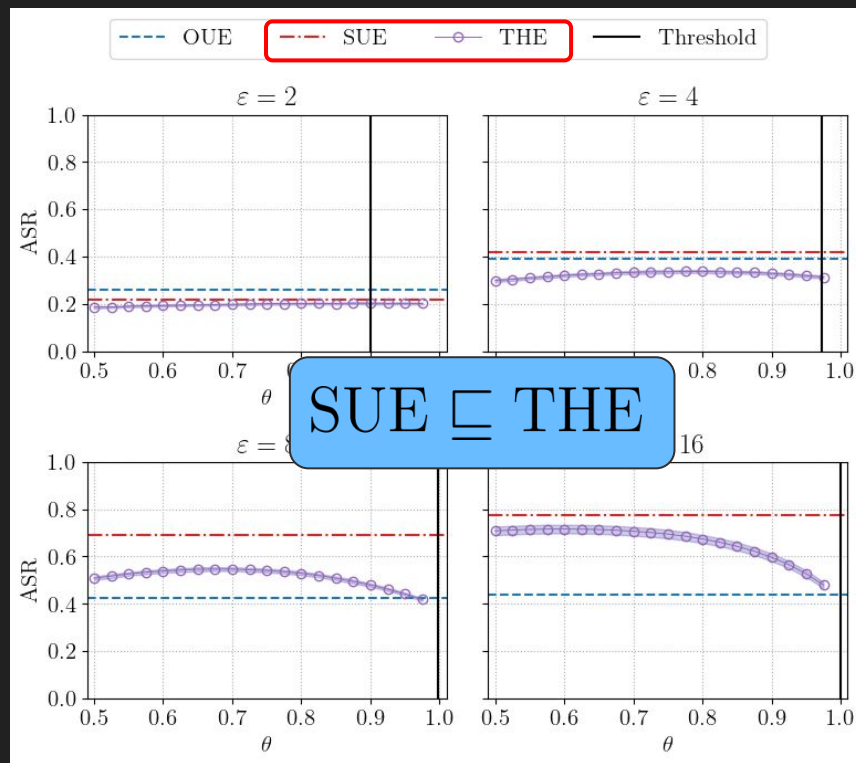


Experiments

Kosarak dataset: Sequences of actions (clicks) performed by users while browsing a website, from a Hungarian news portal.

- 41,270 distinct actions
- 8kk actions performed by 990k users.

Comparison of ASR
Sample of 10,000 actions,
top 10 most frequent actions.

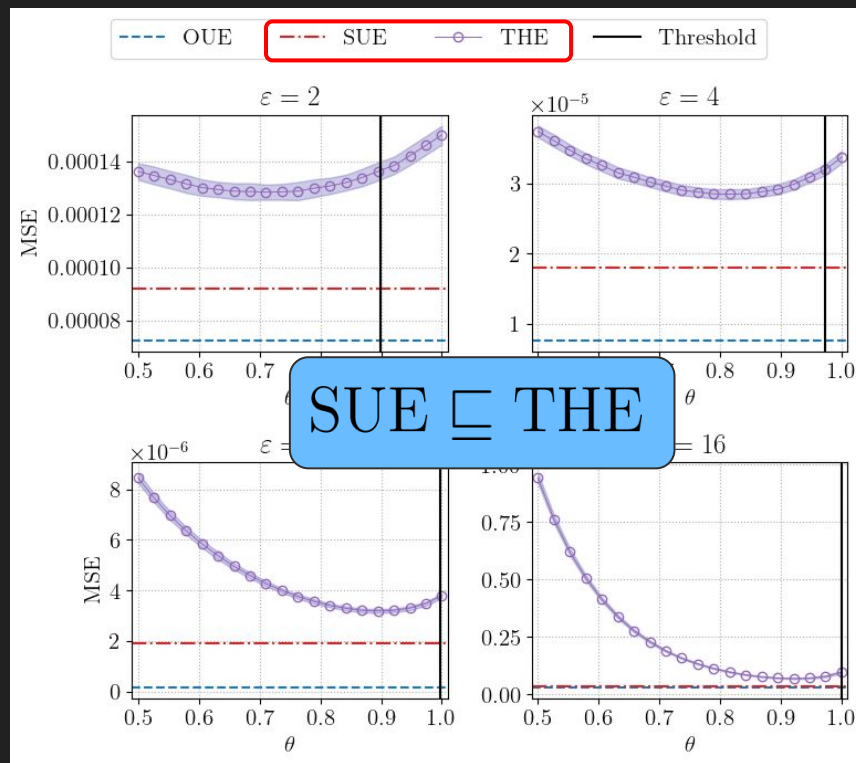


Experiments

Kosarak dataset: Sequences of actions (clicks) performed by users while browsing a website, from a Hungarian news portal.

- 41,270 distinct actions
- 8kk actions performed by 990k users.

Comparison of MSE
Sample of 10,000 actions,
uniformly at random



Limitations & Future work

- Analyze more complex tasks: heavy hitter discovery, multidimensional data.
- Refinement relations for SS and LH.
- Compositional properties of refinement:
 - What happens under sequential composition of a protocol?
- Extension to Metric-LDP (e.g., Geo-indistinguishability)



“That’s all Folks!”

ramon.gonze@inria.fr